

UPDATE REPORT · 18 SEPTEMBER 2026 · 21:25 PT

JARVIS gap report: job/resume semantic matching vs. your spec

This is the exact distance between the 14-page **Semantic Matching Fix** and the running build. Page extraction is treated as already working. This report is about whether comparison is semantically trustworthy. Nothing is marked done without the same Deloitte 359035 live comparison passing all five required tests.

Build: restarted after this update | **Tests:** 188 automated passed | **Live Deloitte semantic proof:** not yet passed

1. The gap in one paragraph

The last live Deloitte run extracted real qualifications — that part held. Comparison did not. JARVIS compared 31 raw lines, including two copies of most requirements, standalone **and/or**, a wage-range paragraph, and “Bachelor's Degree” as a gap even though a bachelor’s was on the resume. Evidence was token overlap: Python matched the word “experience”; Azure/GCP matched consulting/sales bullets; the degree matched data-mining lines. “What matched well?” returned zero strong matches for the wrong reason, and “Why Python?” dumped the partial list. This update replaces that keyword classifier with canonical requirements, fragment/boilerplate filters, a semantic retriever, degree/years logic, and RequirementLookup. That is shipped in code and covered by unit tests on a Deloitte-shaped fixture. It is **not** live-certified on posting 359035 against your actual resume.

2. Scoreboard — you vs. this build

Your requirement	What the build does now	Status
Canonical Requirement objects before compare	CanonicalRequirement stores id, canonical_text, category, required/preferred, min_years, technologies, degree fields, travel %, certs, source provenance.	SHIPPED
Deduplicate by normalized semantic identity	Repeated 4+ / 2+ year lines collapse. Short “Bachelor's Degree” merges into the full technical-degree requirement. Cloud Azure/AWS/GCP alternatives become one requirement.	SHIPPED
Filter compensation / legal / benefits / navigation	Classified COMPENSATION_CONTEXT, LEGAL_CONTEXT, BENEFITS_CONTEXT, NON_REQUIREMENT and excluded from scoring. Wage and annual-incentive paragraphs cannot become gaps.	SHIPPED
Reject fragments (and/or, or, including, etc.)	RequirementFragmentValidator. and/or never appears in missing / partial / biggest gaps in unit tests.	SHIPPED
Split compound requirements as parent + subcomponents	Agentic line stays one parent; orchestration / frameworks become subcomponents, not orphan rows.	SHIPPED
ResumeEvidenceRetriever with minimum threshold	Scores concept hits (python, azure, consulting, ...). Generic tokens (experience, engineer, Herbert, high-growth) have near-zero weight. No forced match.	SHIPPED

Your requirement	What the build does now	Status
No generic-word matches	Azure stack vs consulting/investor-outreach is NOT_FOUND in tests. “experience” cannot cover Python.	SHIPPED
Category-aware matching	Degree looks at education; cloud ignores salesy bullets; travel/sponsorship default to NEEDS_CONFIRMATION unless explicit.	SHIPPED
Degree level vs field vs related-field ambiguity	Bachelor’s in Finance & Economics is PARTIAL/NEEDS_CONFIRMATION with a field-fit reason. It is not “Bachelor’s Degree — not shown.”	SHIPPED
Years from resume chronology, not one bullet	Intervals are unioned from dated roles. COVERED only if timeline ≥ minimum. Unclear dates → PARTIAL, not COVERED.	SHIPPED
Strong-match threshold; zero strong matches allowed	COVERED requires direct concepts, category fit, duration if required, ≥75% score. Empty strong list explains the threshold.	SHIPPED
Partial = related evidence + missing components	Python + ML without verified 2-year timeline is PARTIAL and lists missing duration / leftover concepts.	SHIPPED
“Not shown” is a meaningful label	Cloud becomes “Azure/AWS/GCP agentic AI stack experience”. Degree becomes “Technical degree field (... or related)”.	SHIPPED
Biggest gaps are high-value required items	Ranks required technical / duration / credentials above preferred boilerplate. Fragments and salary cannot rank.	SHIPPED
RequirementLookup for targeted follow-ups	“Why did you mark Python as covered/partial?” resolves one Python item and answers status / requirement / evidence / reason / missing proof. planner_calls=0.	SHIPPED
Store source evidence on each item	Each RequirementItem stores requirement_id, evidence ids/text, status, reason, missing_components, confidence, semantic_similarity, duration_verified.	SHIPPED
Clean HUD format	STRONG MATCHES / PARTIAL MATCHES / NOT SHOWN / NEEDS CONFIRMATION / BIGGEST GAPS. No duplicates, fragments, or wage text in tests.	SHIPPED
Telemetry: raw vs canonical, duplicate groups, per-item scores	Logged on every compare. Follow-up still planner_calls=0.	SHIPPED
Preserve page-extraction / Phase 6 / follow-up / permissions	WEB_DOCUMENT, JSON-LD, chrome filter, RecentAnalysisFollowupResolver, Phase 6 runner unchanged in contract. 188 related tests passed.	SHIPPED
Live Test A — same Deloitte 359035 comparison	Required. Not run on this restarted build.	GAP
Live Test B — targeted Python follow-up	Required after A. Unit-tested; not live-proven.	GAP
Live Test C — “What matched well?”	Required after A. Empty strong list now explains the threshold.	GAP
Live Test D — duplicate counts in logs	Code logs raw/canonical/duplicates_removed. Not captured live.	GAP
Live Test E — forbidden items count=0	Unit tests assert and/or, wage, incentive, chrome UI are absent. Live log not captured.	GAP

3. What I changed in this update

Root cause of the live failure: **JobResumeComparisonEngine._classify** scored requirements by token overlap against any resume line. Distinctive tokens were anything longer than two characters after a small stop list, so “experience”, “development”, and “engineer” counted. There was no canonicalization step, so JSON-LD plus educationRequirements plus repeated qualification blocks produced 31 comparison rows.

Correct path now:

- JobPostingModel still supplies WEB_DOCUMENT requirements (JSON-LD first).
- RequirementContentClassifier drops wage / legal / benefits / nav / fragments.
- canonicalize_requirements() builds CanonicalRequirement objects, merges and/or cloud alternatives, and dedupes semantic twins.
- ResumeEvidenceRetriever parses education / experience / skills and dated intervals.
- classify_requirement() applies category, degree, years, and 0.75 / 0.45 thresholds.
- RequirementLookup answers one stored item for a follow-up utterance.
- HUD prints STRONG / PARTIAL / NOT SHOWN / NEEDS CONFIRMATION / BIGGEST GAPS.

Files added or materially changed:

- **src/agent/phase6/requirements.py** — canonical objects, fragment validator, content filter, dedupe.
- **src/agent/phase6/evidence.py** — ResumeEvidenceRetriever, chronology, degree/years, thresholds.
- **src/agent/phase6/lookup.py** — RequirementLookup + targeted answer format.
- **src/agent/phase6/compare_types.py** — shared ComparisonResult / RequirementItem fields.
- **src/agent/phase6/compare.py** — engine now canonicalizes, retrieves, logs, and formats cleanly.
- **src/agent/phase6/job_posting.py** — validator rejects fragments and compensation before compare.
- **src/agent/corrections.py** — “Why did you mark Python as covered/partial?” is SHOW_EVIDENCE.
- **tests/test_semantic_matching.py** — Deloitte-shaped fixture covering the failure modes from the live run.

4. What the automated Deloitte-shaped fixture now does

This is not your live resume. It is a fixture built from the bad live output so the same failure modes cannot silently return. On that fixture the engine now produces:

- No and/or, wage range, annual incentive, Zoom, or standalone “Bachelor's Degree”.
- Duplicates removed (raw 25 → canonical 12 on the last fixture run; counts are logged).
- Python = PARTIAL with Python/ML evidence and missing 2+ year timeline verification.
- Azure/AWS/GCP stack = NOT_FOUND (consulting/sales bullets do not count).
- Technical degree = PARTIAL because Finance & Economics is shown and field-fit is unclear.
- Travel and sponsorship = NEEDS_CONFIRMATION.

- Agentic frameworks = NOT_FOUND (no LangChain / LangGraph / orchestration evidence).
- “Why did you mark Python...?” returns one Python answer, not the whole partial list.

Your real resume may produce a different mix of COVERED / PARTIAL. That is expected. The fixture only proves the engine no longer invents evidence or scores boilerplate.

5. Required live tests — still yours to run

The spec is explicit: do not declare fixed until the same Deloitte 359035 live comparison passes all five tests below. JARVIS is restarted and waiting.

Test	What you do	Pass looks like	Current
A — same Deloitte posting	Rerun the exact 359035 compare command.	No Chrome UI, no extension labels, no and/or, no salary disclaimer, no duplicates. Evidence is relevant. Degree handled as field-fit, not “degree missing.” Travel/sponsorship in NEEDS_CONFIRMATION if not explicit.	GAP
B — Python follow-up	Ask: “Why did you mark Python as covered/partial?”	One targeted Python answer (status, exact requirement, snippet, reason, missing proof). planner_calls=0. No partial-list dump.	GAP
C — strong matches	Ask: “What matched well?”	Only genuinely strong items with direct evidence — or none, plus a short threshold explanation.	GAP
D — duplicate check	Read the compare logs.	raw_requirement_count, canonical_requirement_count, duplicate_groups, duplicates_removed. Repeated Deloitte lines collapse.	GAP
E — noise check	Inspect HUD + logs.	Forbidden items (and/or, salary disclaimer, annual incentive, browser UI, extension labels) count = 0.	GAP

6. Carried-forward gaps (still true)

Earlier requirement	Code	Live proof
WEB_DOCUMENT-only extraction; no Chrome toolbar as requirements	SHIPPED	PROVEN on prior live Deloitte run — keep watching
JSON-LD JobPosting parsing	SHIPPED	PROVEN on prior live run (structured_jobposting_found=true)
Final answer is the comparison, never “Completed: Compare and list gaps.”	SHIPPED	UNPROVEN on this matching build
HUD detailed list + short TTS	SHIPPED (new section layout)	UNPROVEN
Local analysis follow-ups, planner_calls=0	SHIPPED + Python lookup	UNPROVEN on live Python follow-up
Chrome extension / CDP / vision DOM bridges	Not built	GAP — unchanged from extraction spec

7. Known remaining engineering gaps (honest)

- **No learned embeddings.** “Semantic” here is concept/alias retrieval plus IDF-style generic suppression, not a transformer. Close paraphrases without alias coverage can still miss.
- **Chronology is brittle.** Only obvious “Mon YYYY – Present/Mon YYYY” and year–year ranges are parsed. Unusual resume date formats will keep year requirements in PARTIAL.

- **Related-field degree is a policy call.** Finance & Economics is treated as unclear, not automatically related. Deloitte may disagree either way.
- **Consulting vs delivery consulting.** SBDC / small-business advisory can score as related consulting. That may be PARTIAL rather than a hard miss.
- **Preferred cert laundry lists** are still one requirement if they appear as one bullet. They are not split into individual certs unless named.
- **Live resume text can differ** from PDF extraction (spacing, missing dates). The engine cannot invent dates that the reader did not extract.

8. Definition of done — still not met

Done means the Deloitte comparison is semantically trustworthy on your machine. It must not:

- compare browser UI
- score salary boilerplate
- emit conjunction fragments
- duplicate requirements
- attach unrelated resume bullets as evidence
- answer a specific follow-up with a broad dump

It must produce clean canonical requirements, relevant resume evidence, accurate COVERED / PARTIAL / NOT_FOUND / NEEDS_CONFIRMATION statuses, precise biggest gaps, and targeted evidence follow-ups. Until Tests A–E pass on posting 359035, matching stays **open**.

9. How I will report after every later update

Each JARVIS change set gets a dated file under **docs/reports/** with your requirement, what the build does, and SHIPPED / PARTIAL / UNPROVEN / GAP. Live tests stay GAP until you run them. Code written is not done.

This report's file: docs/reports/JARVIS_GAP_REPORT_2026-09-18_semantic_matching.pdf

Generated 2026-09-18 21:30 PT from the semantic matching update. Automated tests: 188 passed. Live certification: not claimed.